

Orestis Zambounis

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)
Deep Learning, Computer Vision,
Software Engineering, Robotics

Experience

Senior ML Engineer · Red Hat (vLLM) · Zurich, CH · Remote (May 2026 - Present)

Core maintainer of `speculators` in the vllm ecosystem.

Senior ML Engineer / Tech Lead · QSC · Zurich, CH · Remote (Jul 2023 - Apr 2026)

- **Promoted to ML Tech Lead in Dec. 2024**, leading inference strategy, architecture, optimization and a **team of 3**.
- Converted single-process architecture to a stage-parallel pipeline, **doubling** throughput and improving scalability.
- Ported vision ML models to TensorRT and DALI, **tripling** speed and reducing VRAM usage by **15%**.
- Increased system speed by **30%** on resource-constrained hardware through dynamic batch inference.
- Technologies: TensorRT, ONNX, Weights & Biases, Grafana, ROS, Docker, GCP, PyTorch, Python.

Machine Learning Engineer · Seervision (ETHZ Spin-off, acq. by QSC) · Zurich, CH · Remote (Aug 2021 - Jul 2023)

- Optimized the real-time vision pipeline, **doubling** supported video streams per hardware unit.
- Achieved **24%** lower latency, **45%** less VRAM, and **10%** higher accuracy via SOTA adoption.
- Designed, prototyped, tuned, and deployed a face recognition system with precision above **95%**.
- Technologies: ROS, OpenCV, CUDA, PyTorch, TensorFlow, Docker, GitLab CI/CD, GCP, Python, C++.

ML Infrastructure Engineer · benshi.ai (funded by BMGF) · Barcelona, ES · Hybrid (Nov 2020 - Jun 2021)

- Built cloud infrastructure enabling **large-scale** ETL pipelines, data analytics and machine learning applications.
- Technologies: Databricks, PySpark, MLflow, Docker, Kubernetes, Azure, GitHub Actions, Pandas, Python.

Full-Stack Machine Learning Engineer · Self-employed (Feb 2019 - Mar 2020)

- Developed a CNN-based face predictor with an **18%** accuracy improvement, optimized for inference.
- Developed full-stack application with cross-platform frontend and microservice-based cloud architecture.
- Technologies: AWS, React, Flask, PostgreSQL, TensorFlow, scikit-learn, Python.

Control Systems Engineer, Intern · Rapyuta Robotics (ETHZ Spin-off) · Tokyo, JP · On-site (Mar 2016 - Feb 2017)

- Achieved a **55x speedup** of NumPy-heavy simulation iterations and open-sourced the Python package `PyJet`.
- Designed energy estimators using a Kalman Filter (EKF), enhanced tracking controller and performed sensor tests.
- Technologies: ROS, NumPy, SciPy, Python, C++.

Education

MSc Robotics, Systems & Control · 5.25/6.0 · ETH · Zurich, CH (2017 - 2019)

- Developed an online deep learning architecture for object instance prediction, pose estimation, and tracking.
- Showed that an additional depth input channel improved the segmentation accuracy of Mask R-CNN by **31%**.
- Technologies: TensorFlow, Keras, Caffe2, OpenCV, CUDA C/C++, Python.

BSc Mechanical Engineering · 5.51/6.0 · ETH · Zurich, CH (2012 - 2016)

- Developed balancing algorithms for a 6DoF `omnicopter` using non-linear control methods.
- Technologies: MATLAB, Simulink, C++.

Projects

Shop Automation: Self-service IoT system built using Flask, Stripe, Shopify, RasPi, RS-485. (2023 - 2025)

Trap the Cat: Mobile app built with JavaScript, CapacitorJS and Firebase, with **100k+** downloads. (2023 - 2024)

Machine Dreams: Experimental fusion of AI and digital art using GANs, creating surreal NFT artworks. (2022)